

mc Capstone



SCHULICH
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Sponsored By:

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The Problem...

Alberta Health Services (AHS) must audit pharmacies for their vaccine temperature data records based on standards set by the Alberta College of Pharmacists (ACP). These monitoring devices are often:

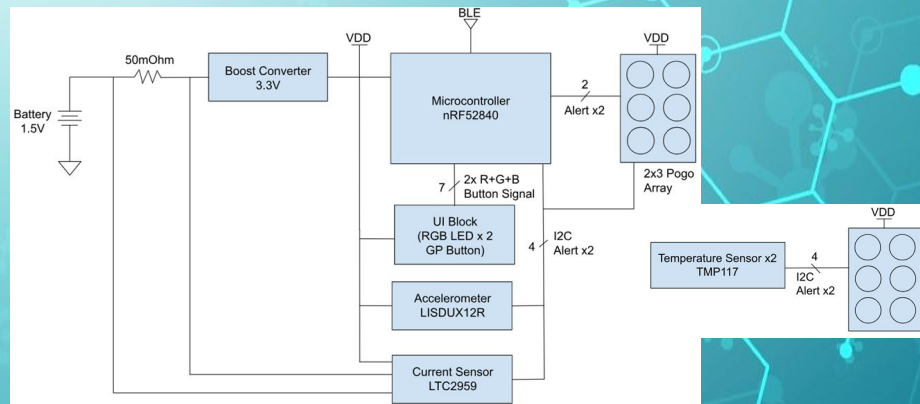
- High cost for wireless (\$300-\$600+)
- Require manual technician calibration
- Large form factor
- Require manual temperature recording
- Short <1-2yr battery life
- None or weak alert system
- Long temperature update interval
- Not sustainable



...Our Solution!

Introducing our product, mcChill:

- Remotely check vaccine temperatures
- Wireless BLE connectivity to a gateway
- Gateway forwards data to the cloud
- Cloud data accessible by pharmacists
- Real-time over/under temp alerts
- Store and forward message system
- Ultra low-power consumption



mcChill (BLE)



mcGate (BLE + LTE)

How do we Compare?

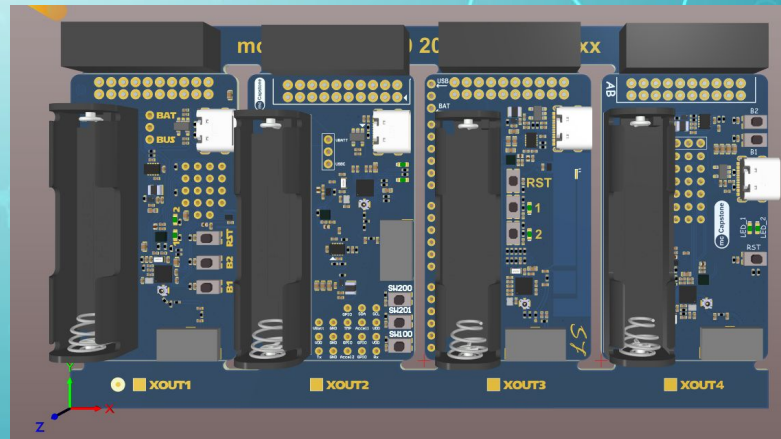
	mcChill	BlueRover
Cost	\$15-\$20 / Unit at scale (\$35-\$50 for Gateway)	\$500+ / Unit
Expected Lifetime	Minimum 10 Years (with battery replacement)	Less than 1 Year (Unit replacement required)
Battery Life	1-2 Years	Less than 1 Year
Battery QTY.	1 x AAA	6 x Li-Ion
Temperature Update Interval	2 Minutes	1 Hour
Communication Protocol	BLE + LTE	LTE
Size	43mm * 65mm	Comparable to tissue box
Accessibility	Access through mcCloud globally with customizable variables and alerting methods (SMS, email)	Proprietary software – SMS and email availability, only modifiable by company.

Initial Development

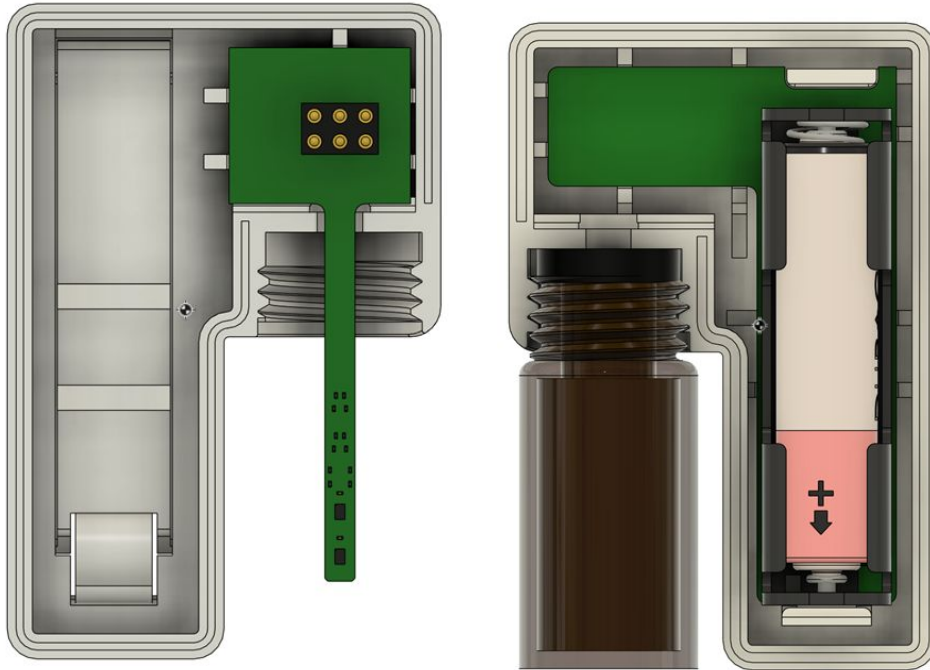
- We knew SW/FW was going to be tough
- Need custom boards for development
- HW team used this to learn Altium

They feature most hardware seen on our final product!

- SW team needed to delve into the intricate and unforgiving world of FW
- Used this opportunity to better understand how to write bulletproof software



Engineering Creativity



We split our hardware into two parts:

- Motherboard:
 - nRF52840 microcontroller
 - LIS2DUX12 accelerometer
 - UI elements
 - Replaceable AAA cell
- Daughterboard:
 - 2x TMP117 temp sensors
 - Interfaces with vial

Both boards interface with each other via a 6 pin pogo array!

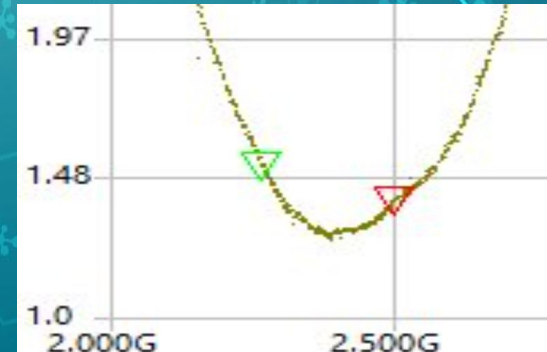
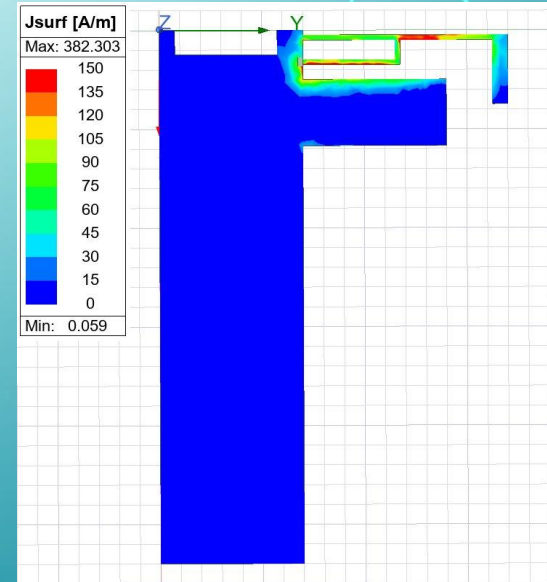
Custom BLE Antenna

Challenges

- Radio signal must get through fridge walls
- Unique shape of PCB requires custom solution

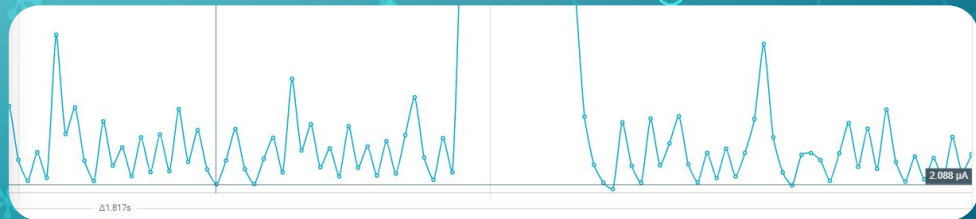
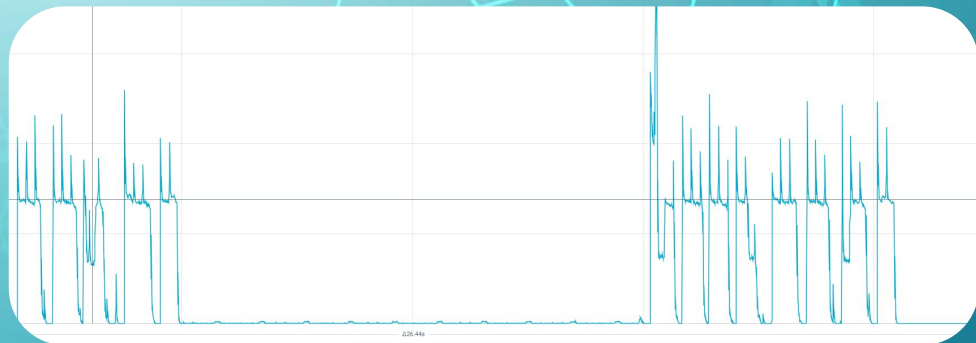
Our Solution

- Designed using Ansys HFSS
- Power outperforms SMT antenna by over 5dB
- Increased range through fridge walls
- 10% reduction in product cost



Power efficiency

The software was designed with power efficiency in mind, adding a deep sleep mode, making our device last long on one battery.



Cloud communication

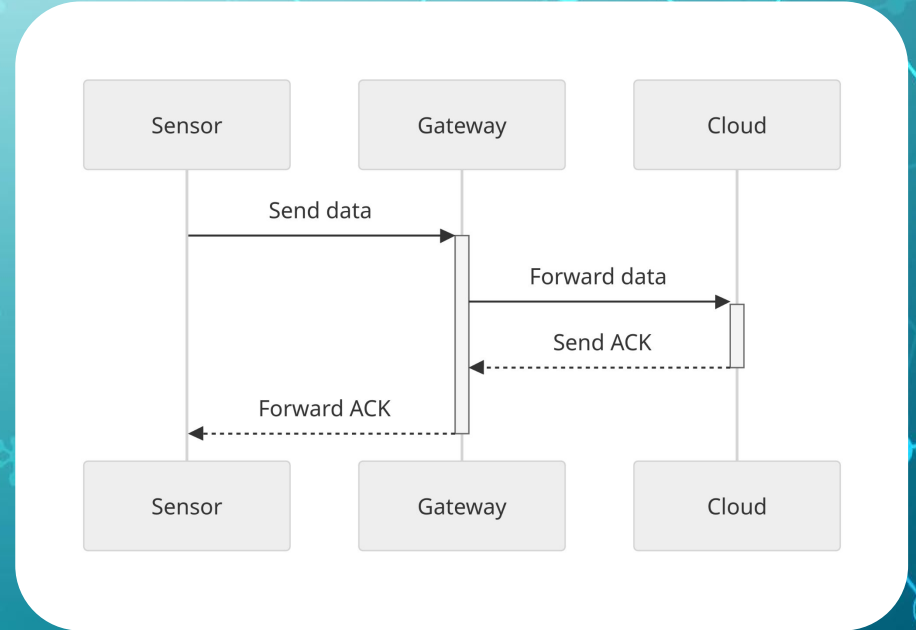
Using BLE data is sent to the LTE-enabled Gateway, enabling cloud communication.

Using the cloud enables remote monitoring of data and an alert system for temperature reading & battery level.



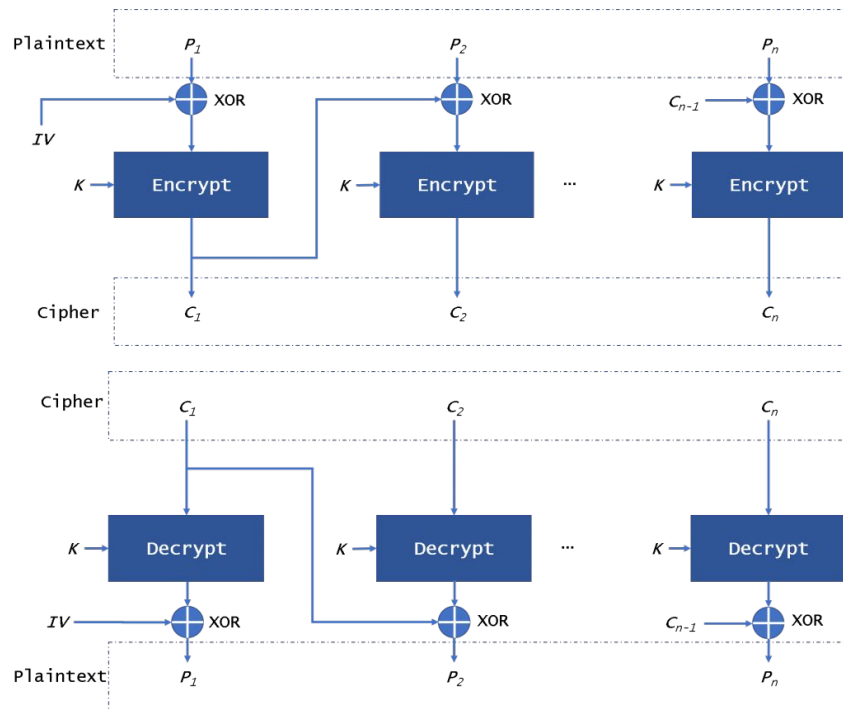
Store-and-forward messaging

Data is stored locally on the device until it is successfully stored in the cloud, ensuring no data is lost.



Encrypted messaging

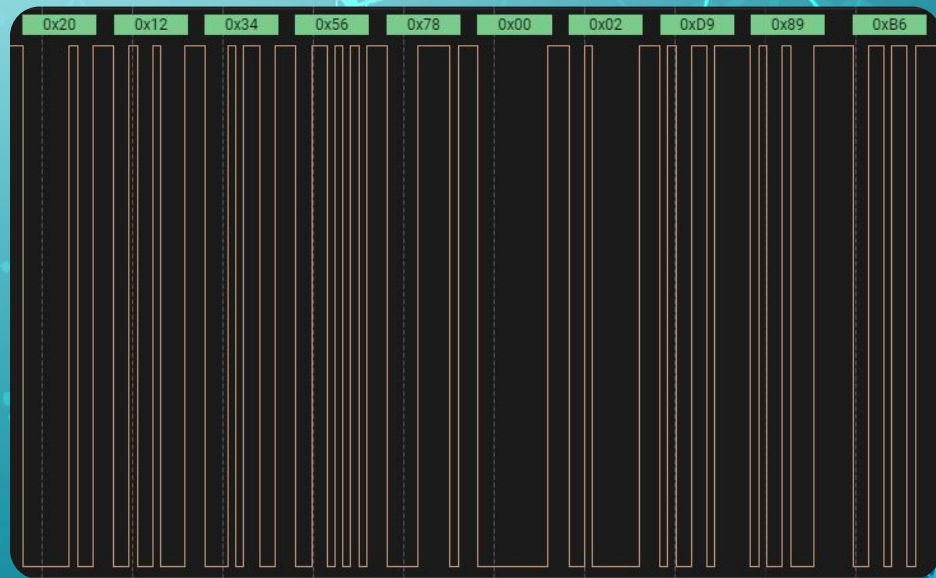
All data is encrypted using local 128-bit keys with AES-128-CBC for private, secure communication, mitigating any MITM or Replay attacks.



Serial Communication

Communication with our on-board sensors utilizes I2C, allowing low-power optimization.

Inter-chip communication is orchestrated through UART using a custom protocol to ensure reliable communication with large buffers and two-way handshakes.

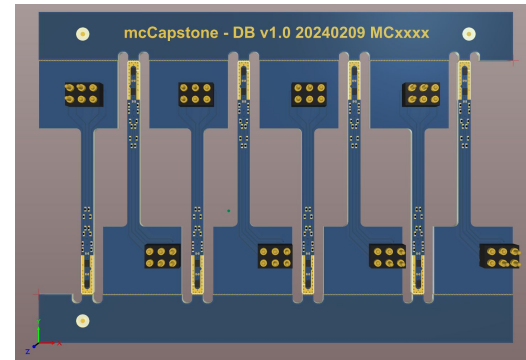
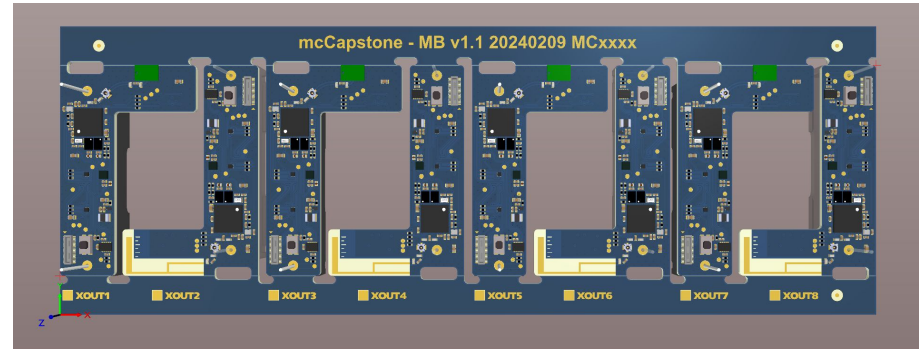


Designed for Manufacturing and Testing

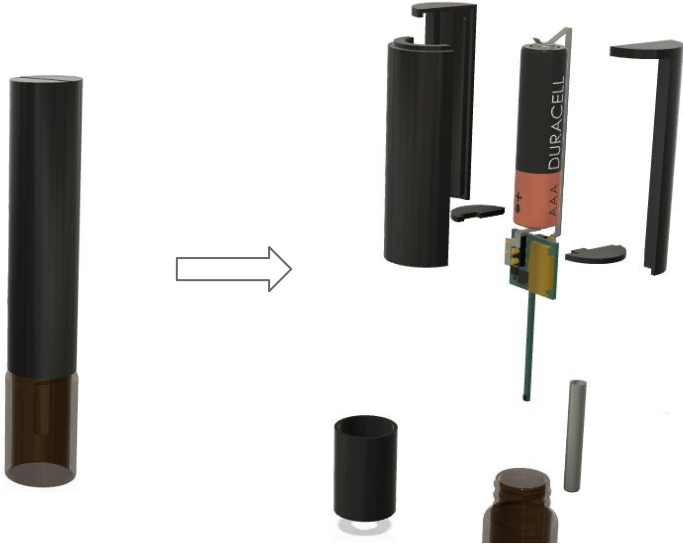
Manufacturing is the most important aspect of a low cost product, often overlooked until the end.

We've considered scale manufacturing since the very beginning, in every single step:

- No silkscreen → precise solder application
- Straight edge board → easy panelization
- Exclusively utilized SMT components → PNP
- Exposed pads for bed-of-nails PRG fixtures
- Proper BGA fan-out minimizes torque in reflow
- Pogo connectors → no 'finicky' assembly
- Unique RGB LED choice → very cost efficient
- 4 layer PCB w/ very common layer stackup
- Reduce heat transfer from probe w/ resistors
- Sectioned paste pads under large components
- Ultrasonic welding → fast, easy, waterproof



Alternate Solutions - Mechanical

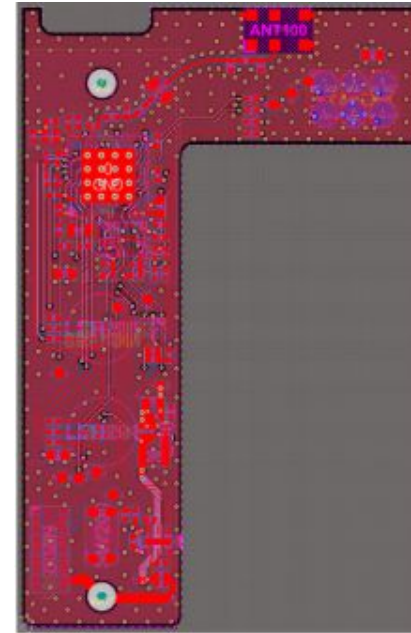


Pros	Cons
Smaller form factor	Off balance - Top heavy
Less material required	Requires expensive PCB fabrication
	Complex antenna design
	More complicated assembly

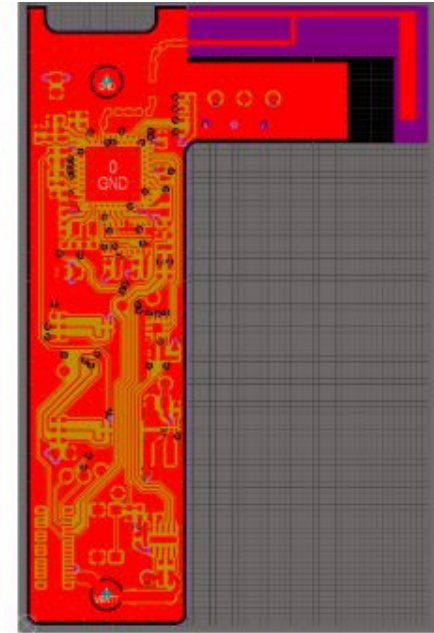
Alternate Solutions - Hardware

Honourable mentions include:

- An entirely LTE device
- Using a thermocouple instead of a digital temp sensor (no calibration - NIST Traceable)
- TMP119 vs TMP117 debate
- nRF vs ESP vs STM
- Li-Ion vs Li/FeS₂ batteries
- PCB vs chip antenna
- PA/LNA required
- Where the product is placed - location in refrigeration unit
- Magnets - attach to metal walls
- Surrogate solution. vs actual vaccine solution
- Induction charging - rechargeable



SMT Antenna

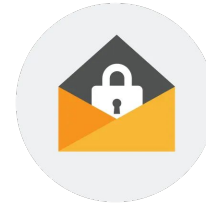
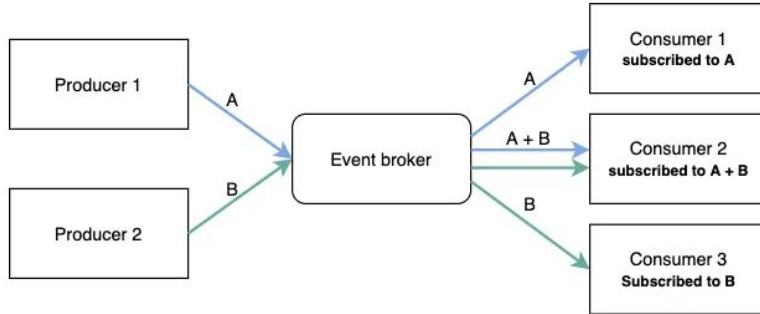


PCB Trace Antenna

Alternate Solutions - Software



- NRF5 SDK vs. Zephyr



- Encrypting messages only vs. BLE encryption
- Event driven vs. Superloop architecture
- Higher level vs. lower level BLE libraries